Lecture 4: Data Collection and Sampling

Hamid Bekamiri
March 27

Textbook: Keller, G. & Gaciu, N. (2020). Statistics for Management and Economics, 2nd ed., Cengage, ISBN: 9781473768260.

Recap - The Big Picture

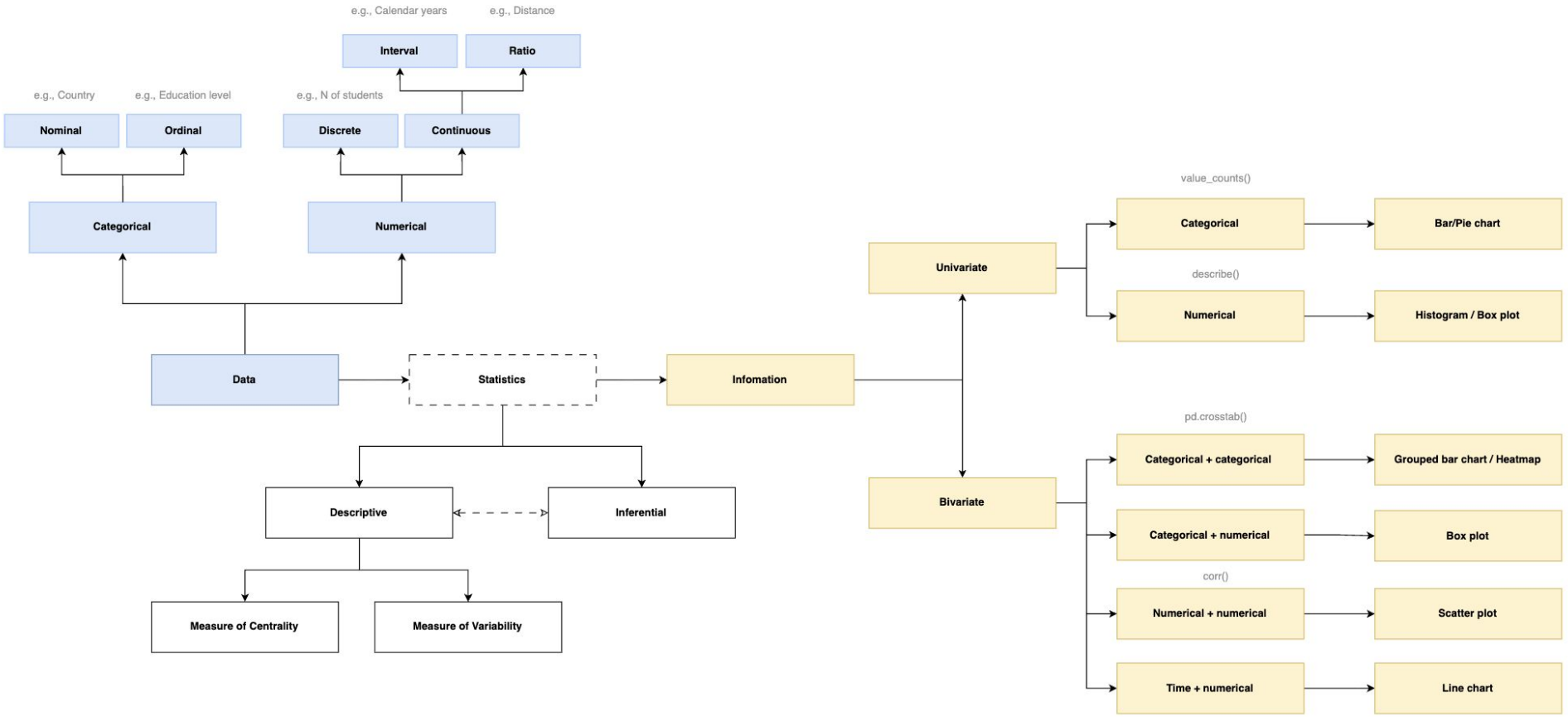
Descriptive Statistics Covered:

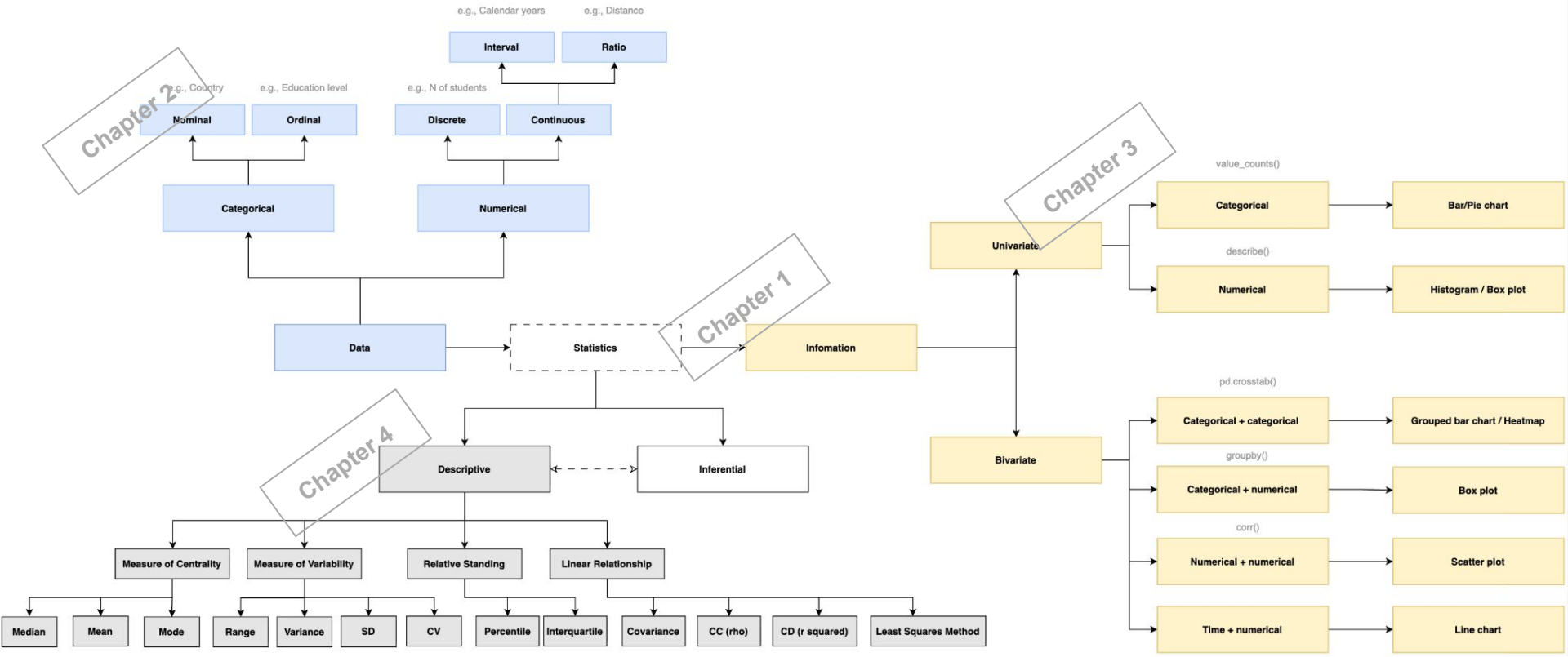
- Chapter 2: Organizing data
- Chapter 3: Summarizing data visually
- Chapter 4: Summarizing data numerically

Next Focus:

- Chapter 5: Data collection and sampling for inference

Goal: Convert raw data into useful business information.


e.g., Calendar years
e.g., Distance
Interval
Ratio
e.g., Country
e.g., Education level
e.g., N of students
Nominal
Ordinal
Discrete
Continuous
Categorical
Numerical
Data
Statistics
Infomation
Descriptive
Inferential
Measure of Centrality
Measure of Variability
Univariate
Bivariate
value_counts()
Categorical
Bar/Pie chart
describe()
Numerical
Histogram / Box plot
pd.crosstab()
Categorical + categorical
Grouped bar chart / Heatmap
Categorical + numerical
Box plot
corr()
Numerical + numerical
Scatter plot
Time + numerical
Line chart


Chapter 2
e.g., Country
e.g., Education level
e.g., N of students
e.g., Calendar years
e.g., Distance
Nominal
Ordinal
Discrete
Continuous
Interval
Ratio
Categorical
Numerical
Data
Statistics
Chapter 1
Infomation
Chapter 3
Univariate
Bivariate
value_counts()
describe()
Categorical
Numerical
Bar/Pie chart
Histogram / Box plot
pd.crosstab()
groupby()
corr()
Categorical + categorical
Categorical + numerical
Numerical + numerical
Time + numerical
Grouped bar chart / Heatmap
Box plot
Scatter plot
Line chart
Chapter 4
Descriptive
Inferential
Measure of Centrality
Measure of Variability
Relative Standing
Linear Relationship
Median
Mean
Mode
Range
Variance
SD
CV
Percentile
Interquartile
Covariance
CC (rho)
CD (r squared)
Least Squares Method

Is this exam hard to pass?

Let's look at the data. You have 5 minutes...

Did last year's class do well?

7, 4, 02, 10, 12, 7, -3, 4, 7, 00, 10, 7, 02, 4, 12, 7, 4, 00,

7, 10, 02, 4, 7, 12

Manual Statistics Practice

A shorter dataset so students can calculate by hand

Use these 5 grades from the original list:

-3, 02, 7, 7, 7

(02 means 2 in the calculations)

Mean	Median	Mode	Variance	Standard deviation

$$\sigma^2 = \frac{\sum f(x-\mu)^2}{n}$$

$$\sigma = \sqrt{\sigma^2}$$

Student Task Slide

Find the mean, median, mode, variance, standard deviation

Data

-3, 02, 7, 7, 7

Sorted:

-3, 2, 7, 7, 7

$$\sigma^2 = \frac{\sum f(x-\mu)^2}{n}$$

$$\sigma = \sqrt{\sigma^2}$$

Suggested board work

Mean = (−3 + 2 + 7 + 7 + 7) / 5 = _____

Median = middle value = _____

Mode = most frequent value = _____

Variance = [(−3 − mean)² + (2 − mean)² + (7 − mean)² + (7 − mean)² + (7 − mean)²] / 5

Standard deviation = √variance = _____

Answer Key

Population formulas used

Final results

Mean	4
Median	7
Mode	7
Variance	16
Std. dev.	4

Working

Mean:
(−3 + 2 + 7 + 7 + 7) / 5 = 20 / 5 = 4

Median and mode:
The sorted data are −3, 2, 7, 7, 7, so the middle value is 7 and the most frequent value is also 7.

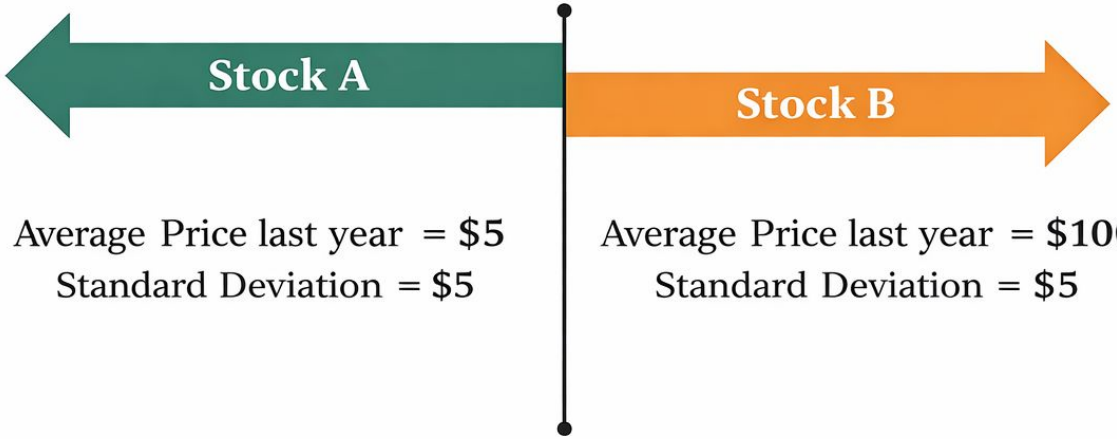
Variance:
$[(-3-4)^2 + (2-4)^2 + (7-4)^2 + (7-4)^2 + (7-4)^2] / 5$
$= (49 + 4 + 9 + 9 + 9) / 5$
$= 80 / 5 = 16$

$$\sigma^2 = \frac{\sum f(x-\mu)^2}{n}$$

Standard deviation:
$\sqrt{16} = 4$

$$\sigma = \sqrt{\sigma^2}$$

Which stock is the best to invest in?



Coefficient of Variance

Stock A: CV = 100%

(5/5*100=100%)

Stock B: CV = 5%

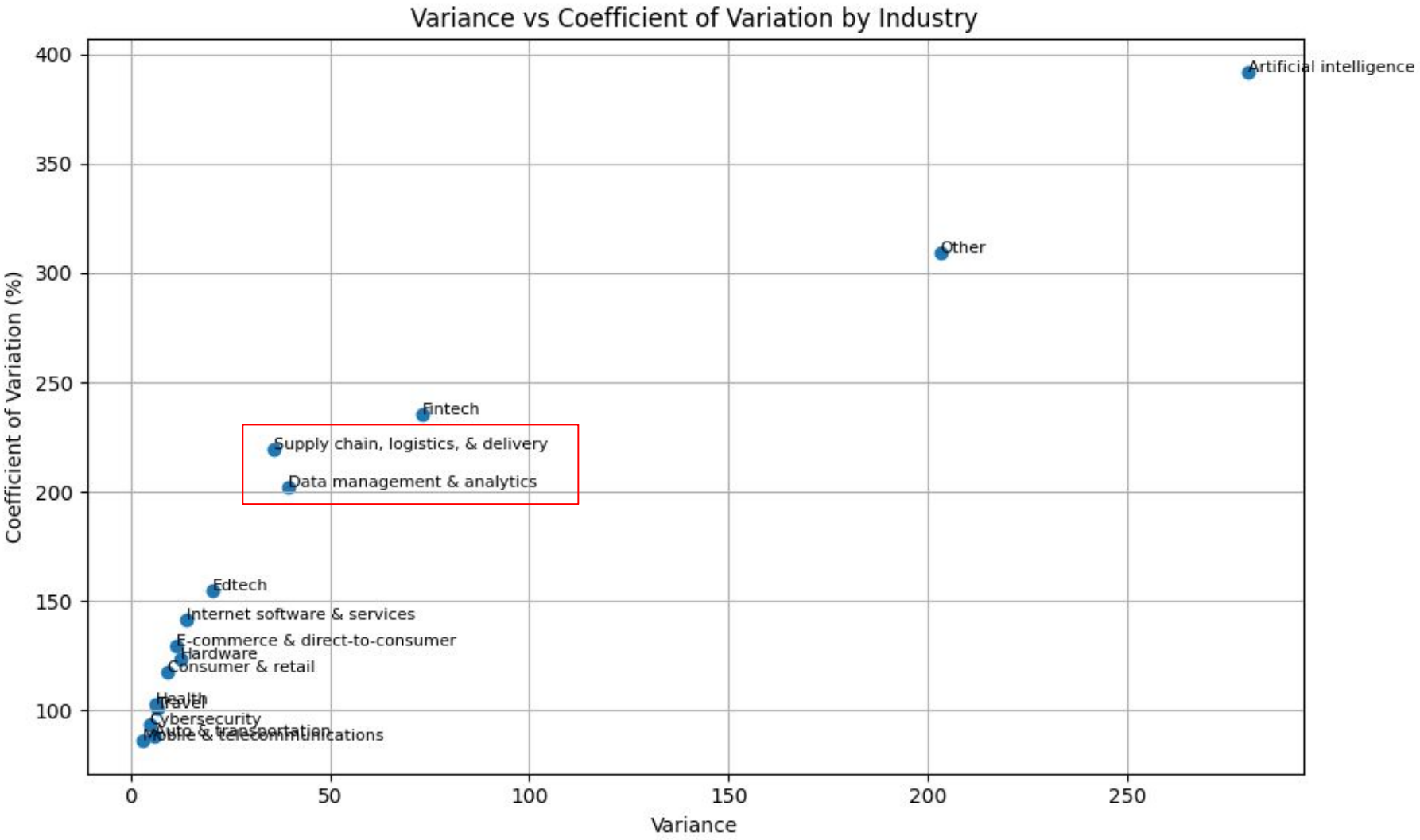
(5/100*100=5%)

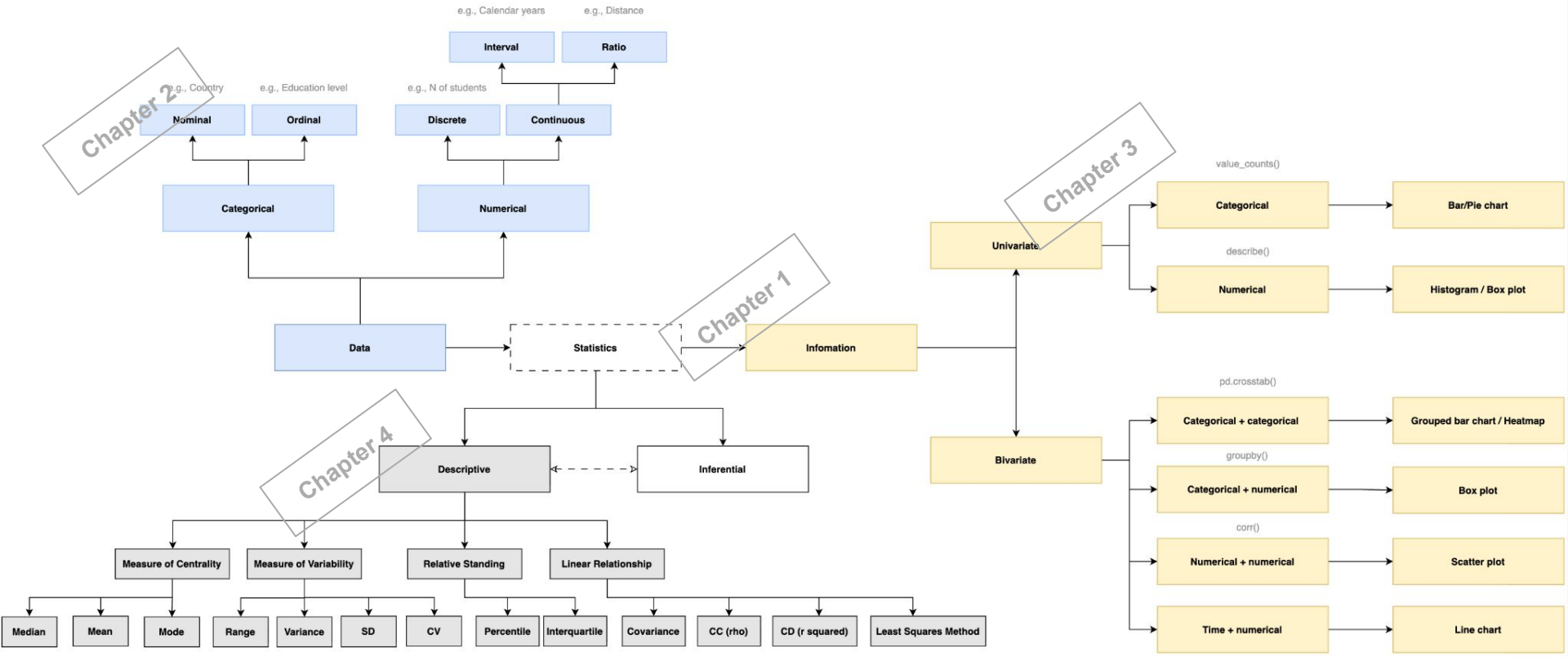
$$CV = \left(\frac{S}{\overline{X}}\right) \cdot 100\%$$

Which industry has greater stability?

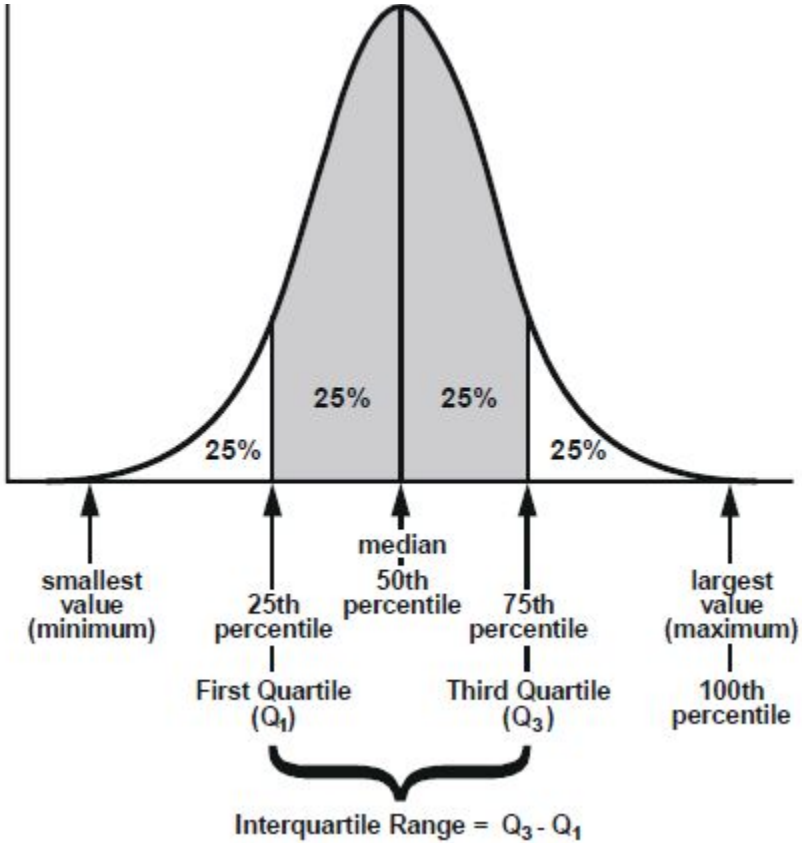
Industry	count	mean	std	min	25%	50%	75%	max
Supply chain, logistics, & delivery	51.0	2.725490	5.976884	1.0	1.0	1.0	2.0	39.0
Data management & analytics	36.0	3.111111	6.296383	1.0	1.0	1.0	3.0	38.0

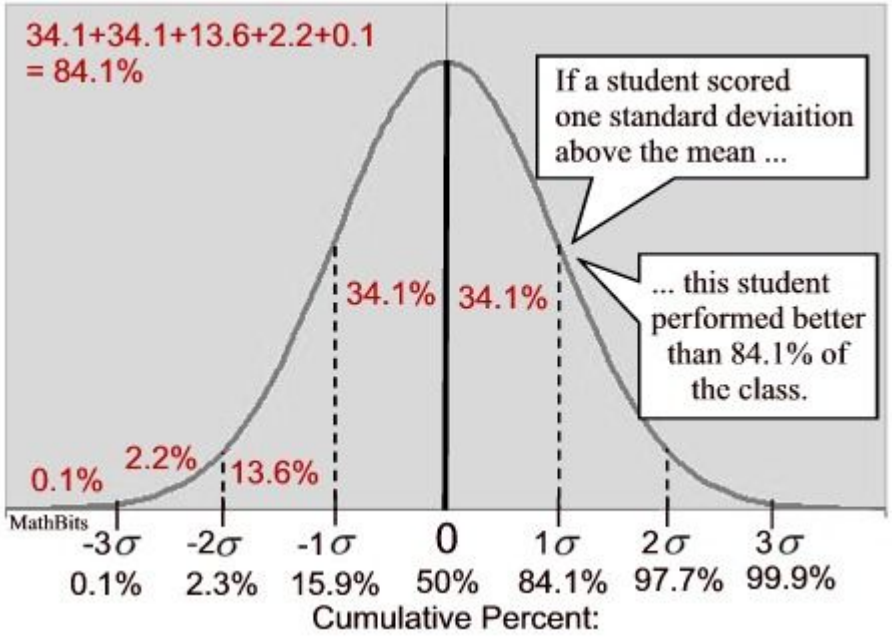
Variance depends on scale; CV standardizes the spread. That is why CV is better for comparing industries with different average valuation levels.

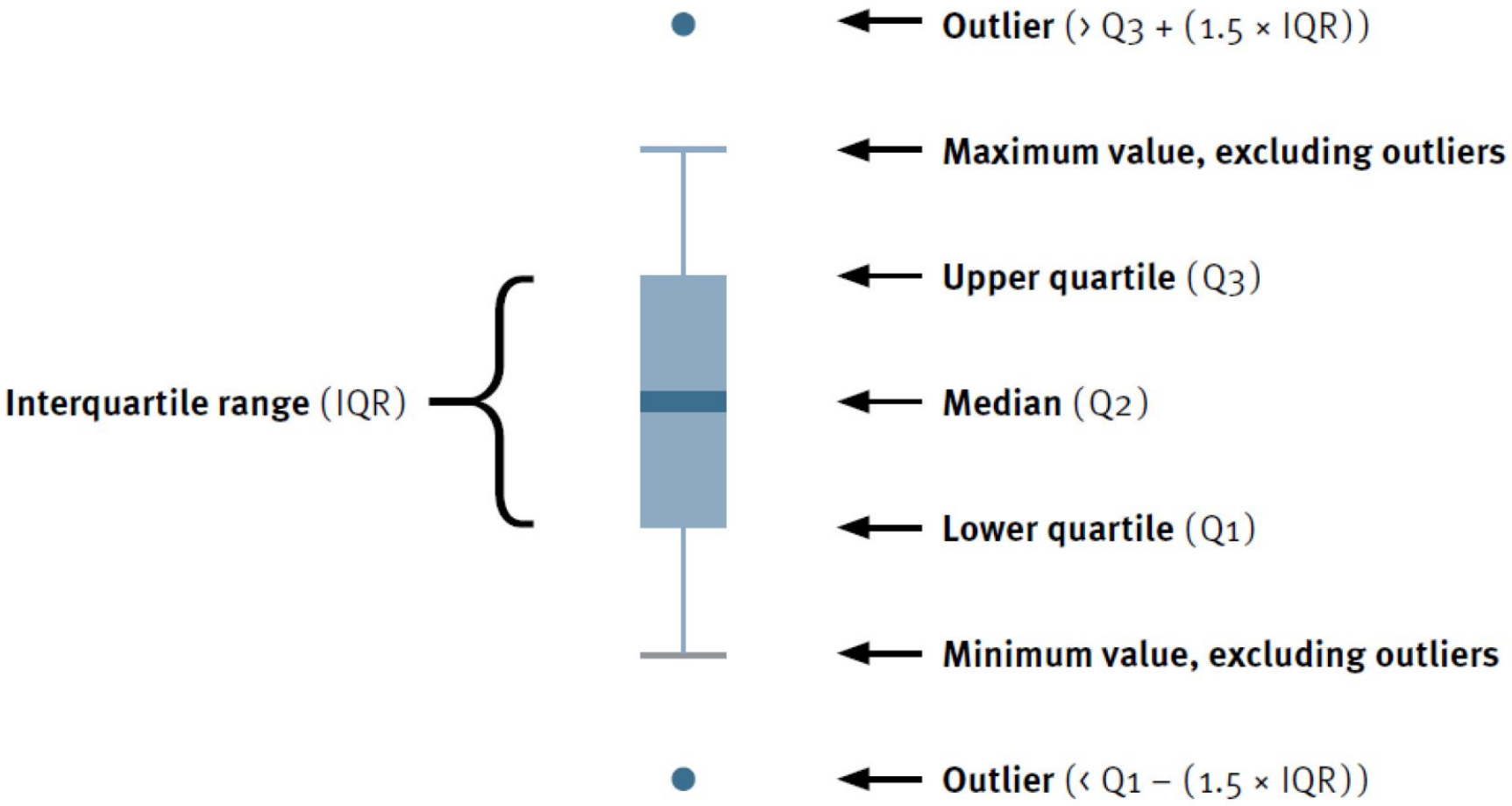

Variance vs Coefficient of Variation by Industry
Coefficient of Variation (%)
Variance
400
350
300
250
200
150
100
0
50
100
150
200
250
Artificial intelligence
Other
Fintech
Supply chain, logistics, & delivery
Data management & analytics
Edtech
Internet software & services
E-commerce & direct-to-consumer
Hardware
Consumer & retail
Health
Travel
Cybersecurity
Auto & transportation
Mobile & telecommunications

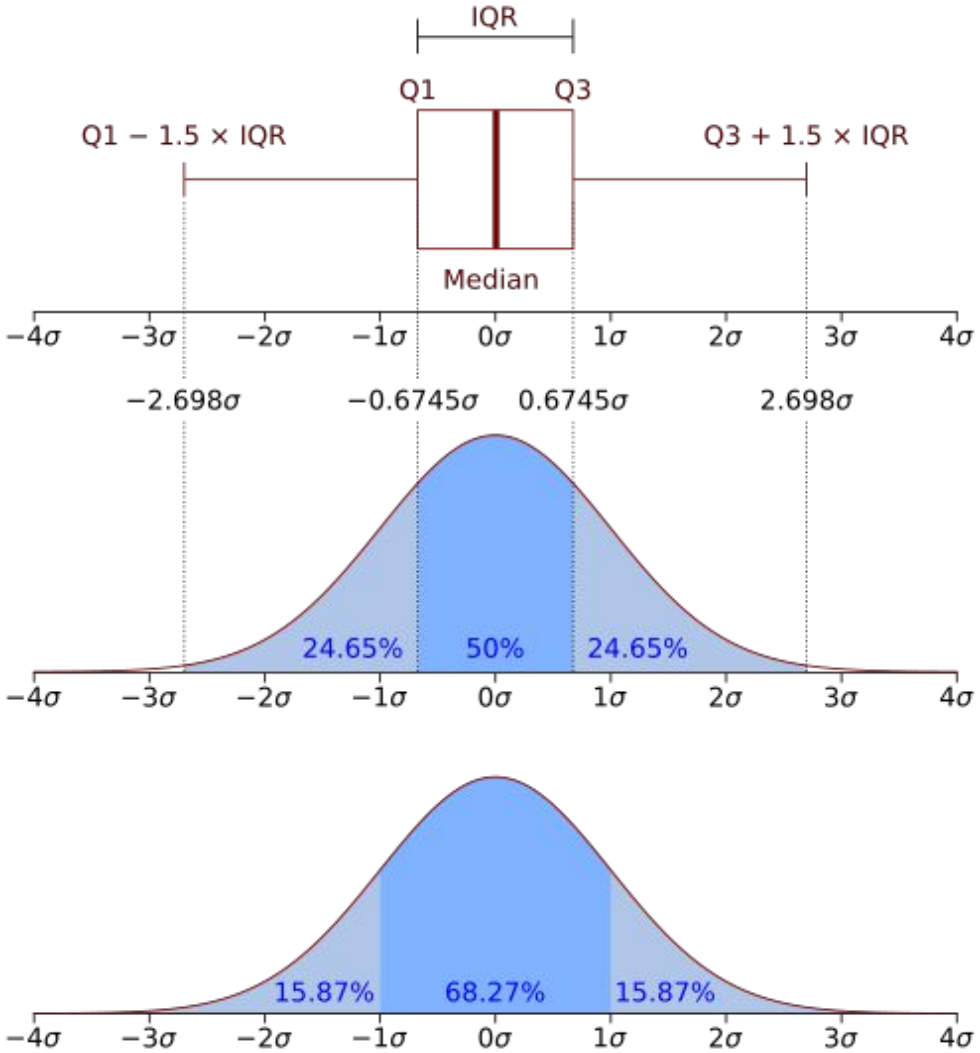

Chapter 2
e.g., Country
e.g., Education level
e.g., N of students
e.g., Calendar years
e.g., Distance
Nominal
Ordinal
Discrete
Continuous
Interval
Ratio
Categorical
Numerical
Data
Statistics
Chapter 1
Infomation
Chapter 3
Univariate
Bivariate
value_counts()
describe()
Categorical
Numerical
Bar/Pie chart
Histogram / Box plot
pd.crosstab()
groupby()
corr()
Categorical + categorical
Categorical + numerical
Numerical + numerical
Time + numerical
Grouped bar chart / Heatmap
Box plot
Scatter plot
Line chart
Chapter 4
Descriptive
Inferential
Measure of Centrality
Measure of Variability
Relative Standing
Linear Relationship
Median
Mean
Mode
Range
Variance
SD
CV
Percentile
Interquartile
Covariance
CC (rho)
CD (r squared)
Least Squares Method

Interquartile range

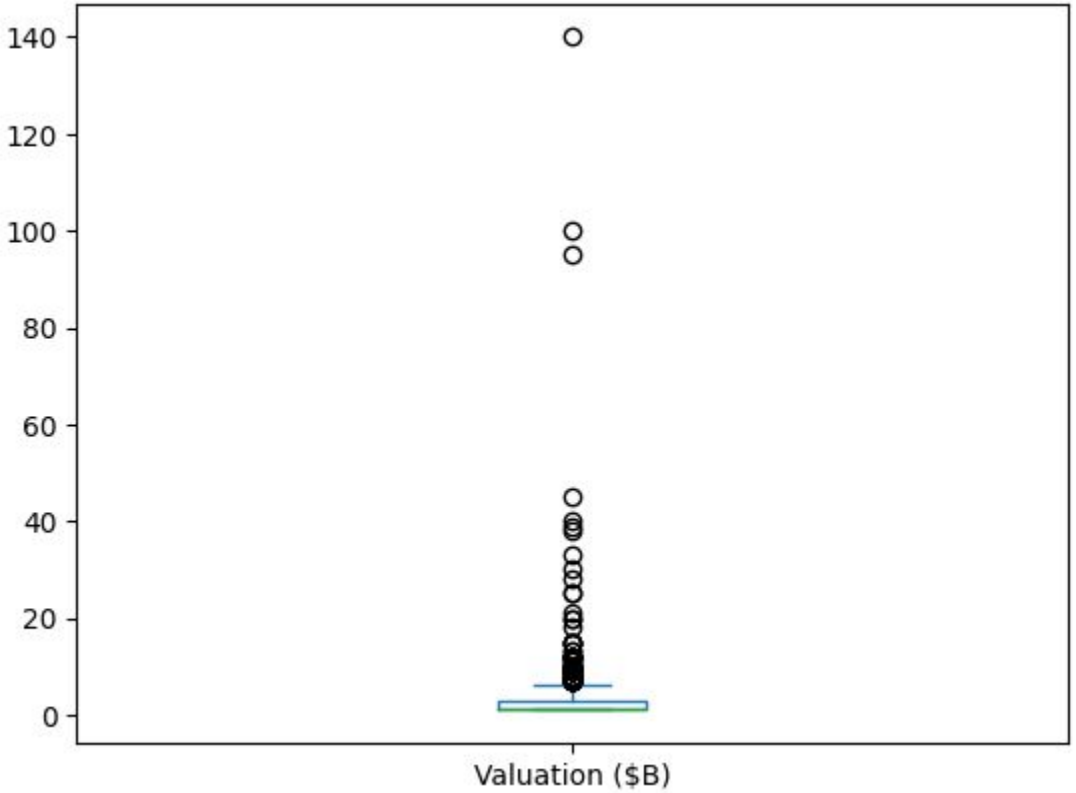


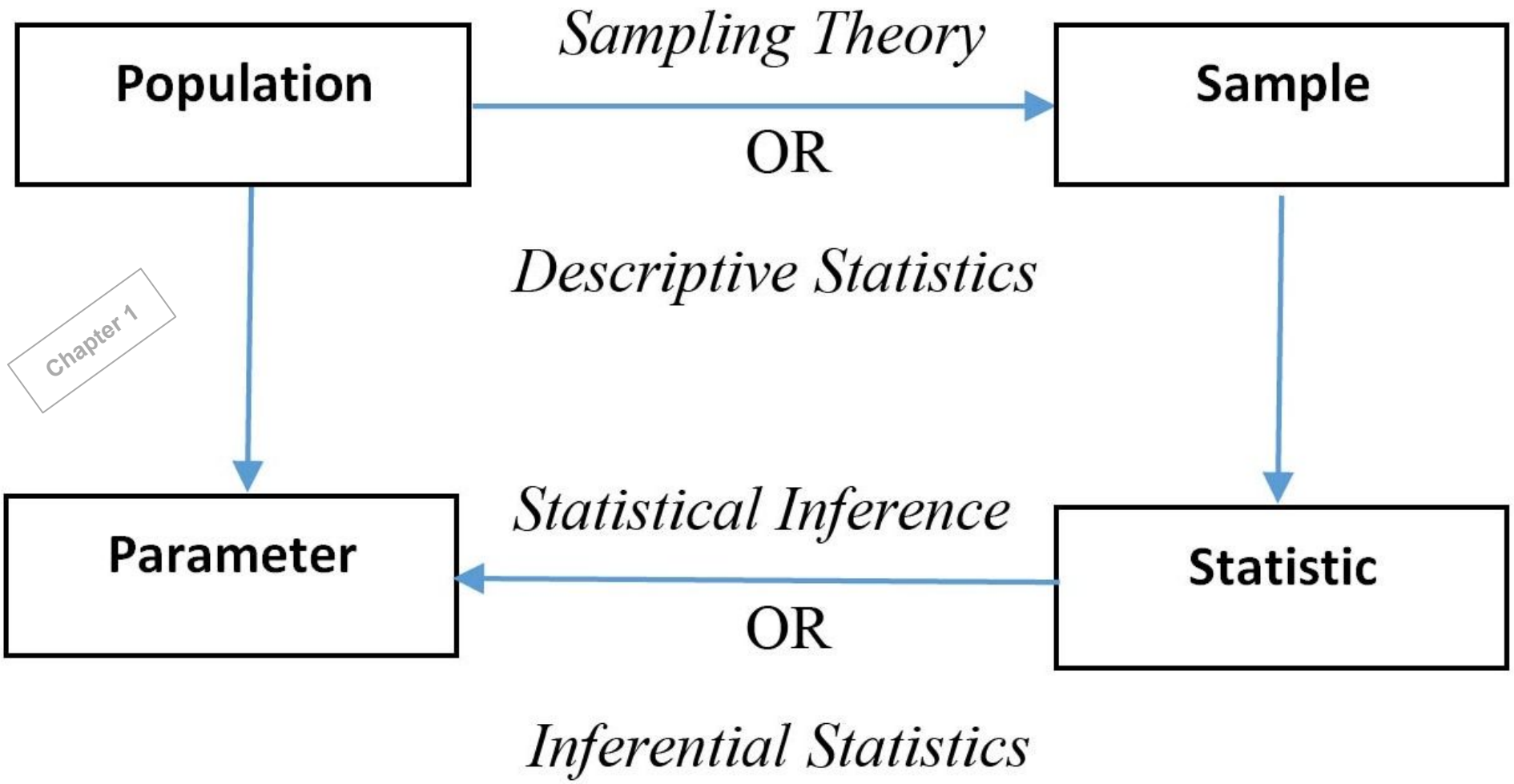



Outlier (› Q3 + (1.5 × IQR))
Maximum value, excluding outliers
Upper quartile (Q3)
Interquartile range (IQR)
Median (Q2)
Lower quartile (Q1)
Minimum value, excluding outliers
Outlier (‹ Q1 – (1.5 × IQR))


IQR
Q1
Q3
Q1 − 1.5 × IQR
Q3 + 1.5 × IQR
Median
−4σ −3σ −2σ −1σ 0σ 1σ 2σ 3σ 4σ
−2.698σ
−0.6745σ
0.6745σ
2.698σ
24.65%
50%
24.65%
−4σ −3σ −2σ −1σ 0σ 1σ 2σ 3σ 4σ
15.87%
68.27%
15.87%
−4σ −3σ −2σ −1σ 0σ 1σ 2σ 3σ 4σ

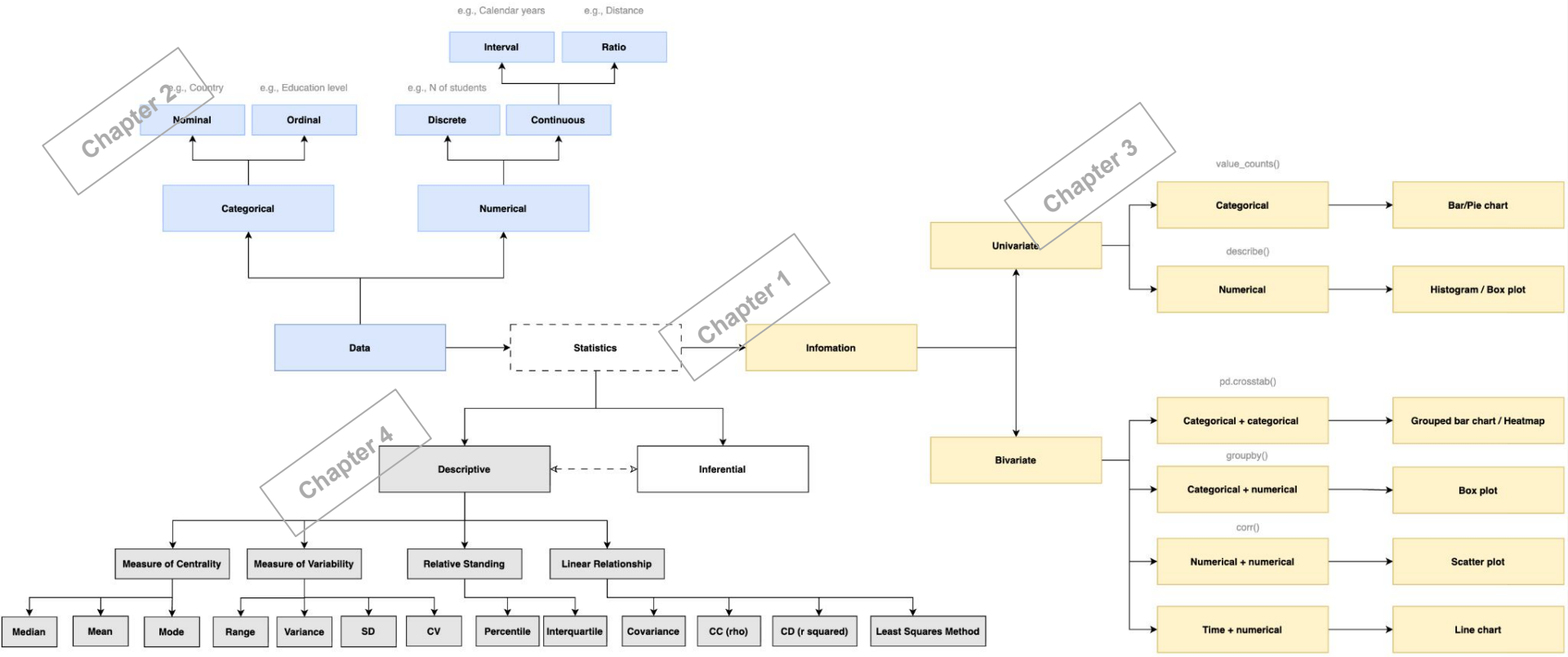
Distribution of Startup Valuations




Population
Sampling Theory
Sample
OR
Descriptive Statistics
Chapter 1
Parameter
Statistical Inference
Statistic
OR
Inferential Statistics

Characteristic	Population Parameter	Sample Statistic
Mean	μ	$\bar{X}$
Median	η	$\tilde{X}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	π (or P)	p (or $\hat{p}$)
Correlation	ρ	r
Regression Coeff.	β	b

Population Variance	Sample Variance
$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$	$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$
$\sigma^2 =$ population variance	$s^2 =$ sample variance
$x_i =$ value of ith element	$x_i =$ value of ith element
$\mu =$ population mean	$\bar{x} =$ sample mean
$N =$ population size	$n =$ sample size


Chapter 2
e.g., Country
e.g., Education level
e.g., N of students
e.g., Calendar years
e.g., Distance
Nominal
Ordinal
Discrete
Continuous
Interval
Ratio
Categorical
Numerical
Data
Statistics
Chapter 1
Infomation
Chapter 3
Univariate
Bivariate
value_counts()
describe()
Categorical
Numerical
Bar/Pie chart
Histogram / Box plot
pd.crosstab()
groupby()
corr()
Categorical + categorical
Categorical + numerical
Numerical + numerical
Time + numerical
Grouped bar chart / Heatmap
Box plot
Scatter plot
Line chart
Chapter 4
Descriptive
Inferential
Measure of Centrality
Measure of Variability
Relative Standing
Linear Relationship
Median
Mean
Mode
Range
Variance
SD
CV
Percentile
Interquartile
Covariance
CC (rho)
CD (r squared)
Least Squares Method

Data and student tasks

Since the dataset is small, you can calculate statistics manually.

Dataset

Student	Hours studied (x)	Grade (y)
1	1	-3
2	2	2
3	3	7
4	4	7
5	5	7

The mean, median, and mode have already been calculated.

Calculate manually

Sample variance and sample standard deviation of the grades

Covariance between study hours and grades

Correlation (r) and coefficient of determination (R^2)

Least squares regression line: $\hat{y} = a + bx$

Suggested order: first calculate grade summaries, then use the x–y table for covariance, correlation, and the least squares line.

Teacher check: grades only

Use 02 = 2 and sort the grades first.

Sorted grades

-3, 2, 7, 7, 7

Mean

(−3 + 2 + 7 + 7 + 7) / 5 = 20 / 5 = 4

Median

Middle value in the sorted list = 7

Mode

Most frequent value = 7

Sample variance and standard deviation

Grade y	$y - \bar{y}$	$(y - \bar{y})^2$
-3		
2		
7		
7		
7		
Total		

Variance = ?

Std. dev. = √? = ?

Optional population results
Variance = ?
Std. dev. ≈ ?

Teacher check: grades only

Use 02 = 2 and sort the grades first.

Sorted grades

-3, 2, 7, 7, 7

Mean

(−3 + 2 + 7 + 7 + 7) / 5 = 20 / 5 = 4

Median

Middle value in the sorted list = 7

Mode

Most frequent value = 7

Sample variance and standard deviation

Grade y	$y - \bar{y}$	$(y - \bar{y})^2$
-3	-7	49
2	-2	4
7	3	9
7	3	9
7	3	9
Total		80

Variance = 80 / 4 = 20

Std. dev. = √20 = 4.47

Optional population results
Variance = 16
Std. dev. ≈ 4

Worked table for covariance, correlation, and slope

The same table helps with covariance, correlation, and the least squares line.

x	y	x − x̄	y − ȳ	(x − x̄)(y − ȳ)	(x − x̄)²	(y − ȳ)²
1	-3					
2	2					
3	7					
4	7					
5	7					
Total				25	10	80

$$Correlation = \frac{Cov\ (x, y)}{\sigma x * \sigma y}$$

x̄ = ?
ȳ = ?

Σ(x − x̄)(y − ȳ) = ?

Σ(x − x̄)² = ? and Σ(y − ȳ)² = ?

$$m = \frac{\text{sum of the terms}}{\text{number of terms}}$$

$$cov_{x,y} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Worked table for covariance, correlation, and slope

The same table helps with covariance, correlation, and the least squares line.

x	y	x − x̄	y − ȳ	(x − x̄)(y − ȳ)	(x − x̄)²	(y − ȳ)²
1	-3	-2	-7	14	4	49
2	2	-1	-2	2	1	4
3	7	0	3	0	0	9
4	7	1	3	3	1	9
5	7	2	3	6	4	9
Total				25	10	80

x̄ = 3
ȳ = 4

Σ(x − x̄)(y − ȳ) = 25

Σ(x − x̄)² = 10 and Σ(y − ȳ)² = 80

$$m = \frac{\text{sum of the terms}}{\text{number of terms}}$$

$$cov_{x,y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Covariance, correlation, and R^2

Covariance

Population covariance = 25 / 5 = 5

Optional sample covariance = 25 / 4 = 6.25

Positive covariance means grades tend to rise as study hours rise.

Correlation

$r = 25 / \sqrt{(10 \times 80)}$

$r \approx 0.884$

The value is close to 1, so the linear relationship is strong and positive.

Coefficient of determination

$R^2 = r^2 = (0.884)^2$

$R^2 \approx 0.781$

About 78% of the variation in grades is explained by study hours in this tiny dataset.

Least squares line method

Find the slope first, then the intercept.

Step 1: slope (b)

$$b = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)} = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

b = 25 / 10 = 2.5

Step 2: intercept (a)

$$a = \bar{y} - b\,\bar{x}$$

a = 4 − (2.5 × 3) = −3.5

Final regression line

$\hat{y} = a + bx$

$\hat{y} = -3.5 + 2.5x$

Quick check
At $x = 3$ hours, $\hat{y} = -3.5 + 2.5(3) = 4$
The line passes through $(\bar{x}, \bar{y}) = (3, 4)$.

Example prediction
For 4 study hours: $\hat{y} = -3.5 + 2.5(4) = 6.5$

Least Square Method – Key Formulas

🔹 **Slope (b)**

$$b = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)} = \frac{\sum(x - \bar{x})(y - \bar{y})}{\sum(x - \bar{x})^2}$$

🔹 **Covariance**

$$\mathrm{Cov}(x, y) = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

🔹 **Variance**

$$\mathrm{Var}(x) = \frac{\sum(x - \bar{x})^2}{n - 1}$$

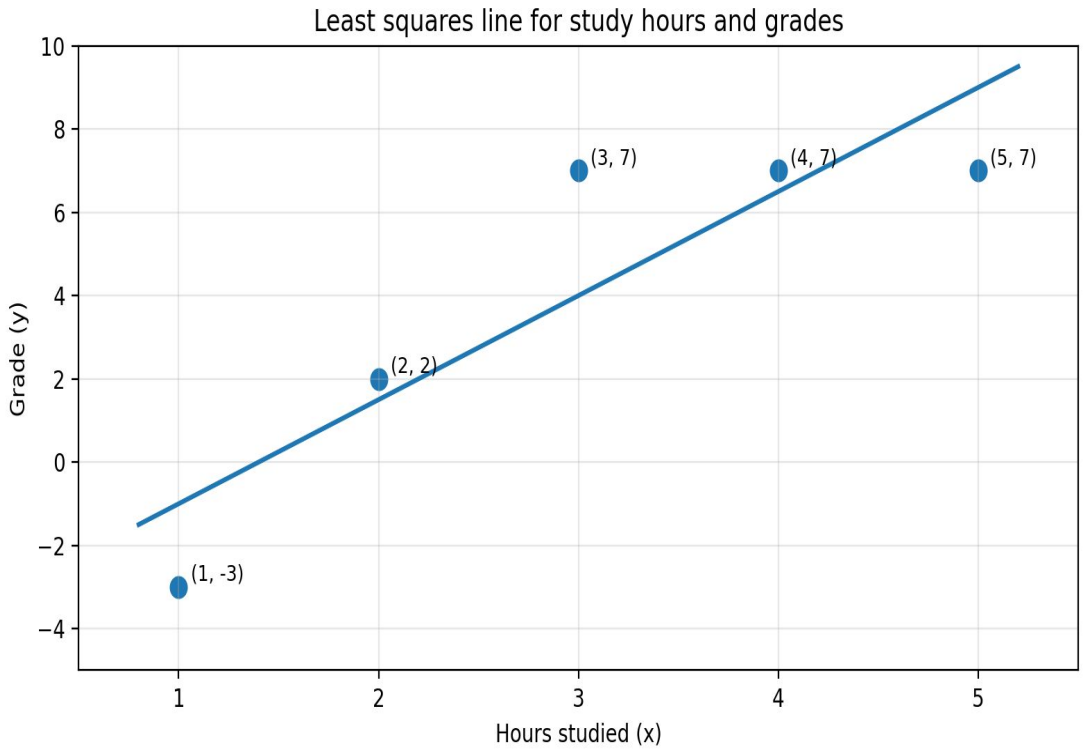
One-slide summary

Use these final answers to check the manual work:

Measure	Answer
Mean of grades	4
Median of grades	7
Mode of grades	7
Population variance of grades	16
Population standard deviation of grades	4
Population covariance	5
Correlation r	0.884
Coefficient of determination R^2	0.781
Least squares line	$\hat{y} = -3.5 + 2.5x$

More study hours are associated with higher grades in this example.

Least squares line: plot, slope, and intercept



Regression line

$\hat{y} = a + bx$

$\hat{y} = -3.5 + 2.5x$

Slope (b) = 2.5

For each extra study hour, the predicted grade
increases by 2.5 points.

Intercept (a) = -3.5

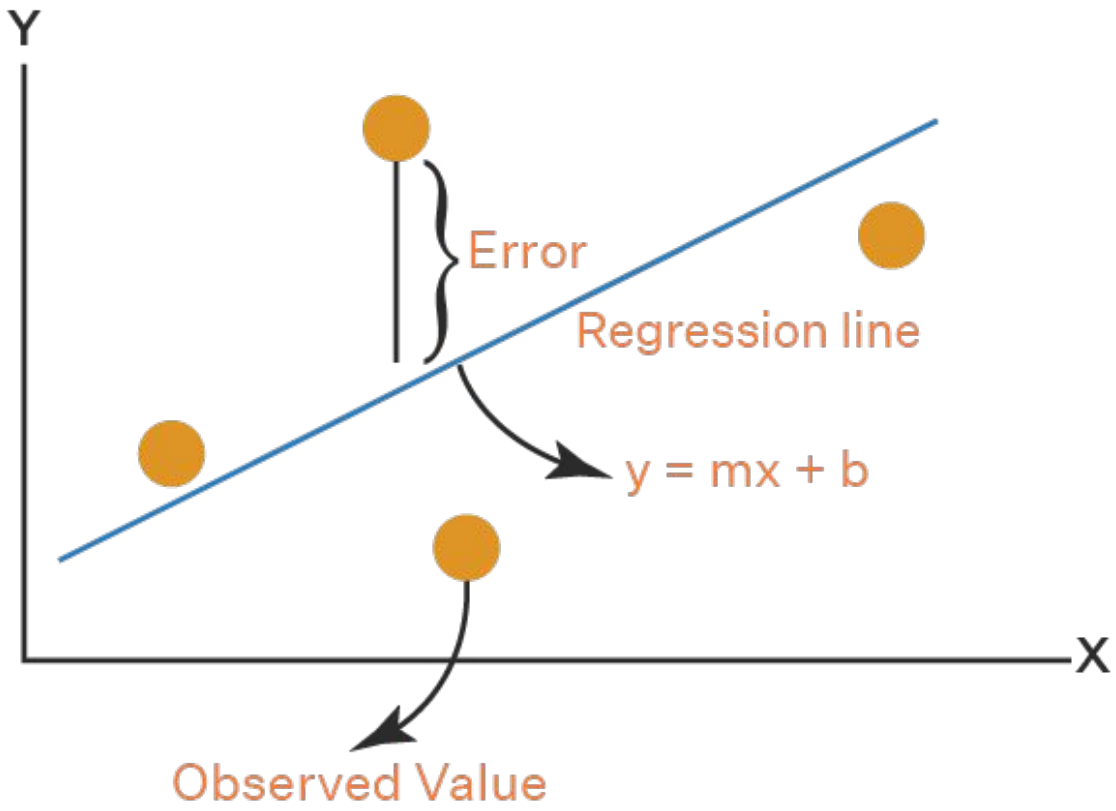
This is the predicted grade when x = 0 hours.

Example prediction

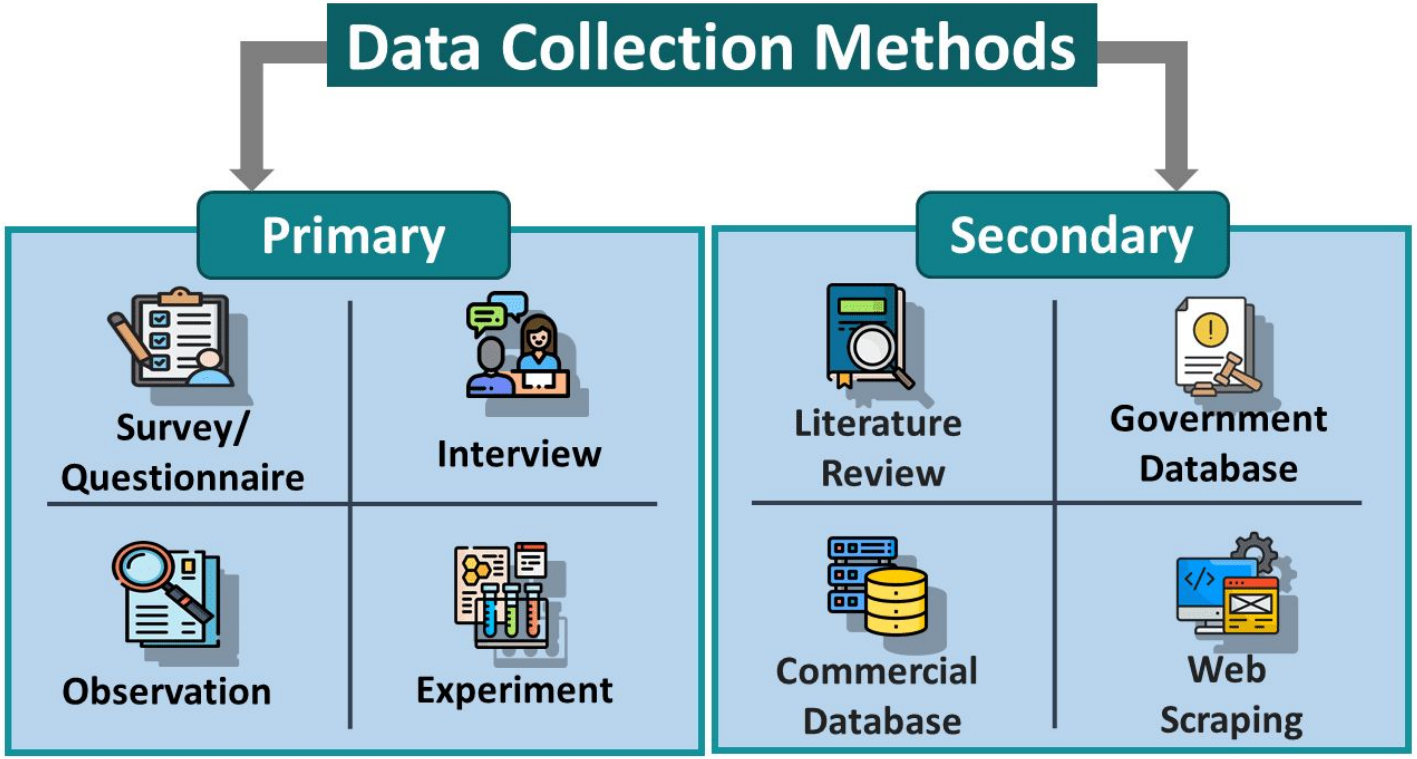
At x = 4: $\hat{y} = -3.5 + 2.5(4) = 6.5$

The least squares line is the straight line that best fits the data overall.

Least Square Method



Data Collection Techniques



Data Collection Techniques

Method	One-sentence explanation	Example
Survey / Questionnaire	Collects standardized responses from many people using the same set of questions.	A university sends an online survey to 500 students about study habits.
Interview	Gathers detailed information by asking individuals questions directly.	A researcher interviews nurses about patient-care challenges.
Observation	Collects data by watching and recording behaviors or events as they occur.	A teacher observes how students interact during group work.
Experiment	Tests cause-and-effect by changing one factor and measuring its impact.	A company changes a website button color to see whether clicks increase.
Literature Review	Uses existing books, articles, and previous studies to gather information on a topic.	A student reviews research articles on social media and mental health.
Government Database	Uses official data published by public institutions.	A researcher uses census data from a national statistics office.
Commercial Database	Uses organized data provided by private companies, often through paid access.	A business analyst uses Bloomberg or Statista to study market trends.
Web Scraping	Extracts data automatically from websites using software tools or scripts.	A researcher collects product prices from e-commerce websites.

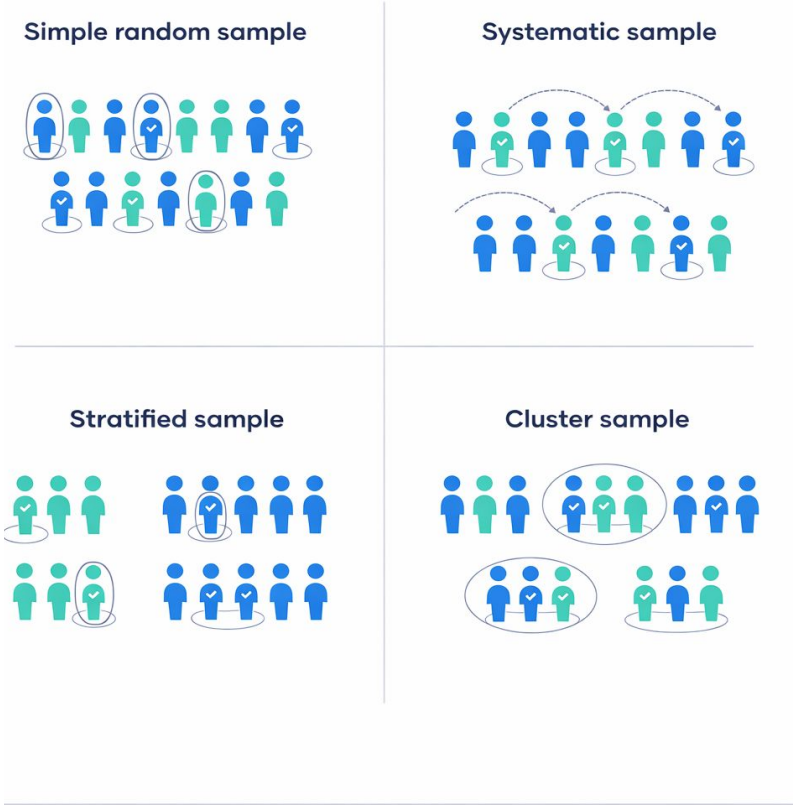
Sampling Techniques

Simple random sample: picks people purely at random

Systematic sample: picks every k-th person after a random start

Stratified sample: splits the population into groups and samples from each group

Cluster sample: splits it into groups but randomly selects whole groups



Sampling Techniques

Sampling method	AAU Business School example
Simple random sample	From the full list of AAU Business School students, randomly select **100 students** to answer a survey about satisfaction with digital learning tools.
Systematic sample	From an alphabetical list of AAU Business School students, randomly choose a starting point, then select **every 20th student** to participate in a study on study habits.
Stratified sample	Divide AAU Business School students into groups such as **Bachelor, Master, and PhD**, then randomly sample students from **each group** to ensure all study levels are represented in a survey on career expectations.
Cluster sample	Treat each **class section or course group** as a cluster, randomly choose a few classes, and survey **all students in those selected classes** about teaching quality.

Sampling and Non-Sampling Errors

Error Category	Specific Type	What It Means (Simply)
Sampling Error *(Flaws in WHO is selected)*	**Population Specification**	Targeting the wrong overall group. Forgetting to exclude non-citizens who cannot legally vote
	Sample Frame	Working from an incomplete list. Missing the hundreds of thousands of Danes who have *navne- og adressebeskyttelse* (protected/unlisted numbers).
	Selection	Biased picking or self-selection. *Politiken* hosts an online poll; mostly left-wing readers choose to participate, falsely predicting a massive Red Bloc landslide.
Non-Sampling Error *(Flaws in the PROCESS)*	**Measurement / Response**	Dishonest answers or bad questions. A "shy voter" feels embarrassed to support a fringe party
	Non-Response	Selected people refuse to participate. Rural retirees pick up the pollster's calls, but busy parents in the Capital Region ignore them, leaving a massive gap in the data.
	Processing	Clerical mistakes or typos.

vevox

Paused
00:09

0 / 0

How do you feel right now when you hear the word "Statistics"?

Bring on the numbers! 📊

0%

I'm cautiously optimistic. 🤔

0%

Slightly nervous, to be honest. 😅

0%

Is it too late to leave the room? 🏃💨

0%

vevox

10 / 10

How do you feel right now when you hear the word "Statistics"?

Bring on the numbers! 📊

60%

I'm cautiously optimistic. 🤔

40%

Slightly nervous, to be honest. 😅

0%

Is it too late to leave the room? 🏃💨

0%

Paused

00:06